

28. Answer: A

$$\begin{aligned}\text{Energy of the photon } \Delta E &= \frac{hc}{\lambda} \\ &= \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{633 \times 10^{-9}} \\ &= 3.1422 \times 10^{-19} \text{ J} \\ &= \frac{3.1422 \times 10^{-19}}{1.6 \times 10^{-19}} \text{ eV} \\ &= 1.96 \text{ eV}\end{aligned}$$

Hence the transition is from W $\rightarrow$ X ($20.66 - 18.70 = 1.96 \text{ eV}$)

29. Answer: C